

Nervous System Health and Neurotechnology

Brain, spinal cord, peripheral nerve, neuromuscular, autonomic, sensory-motor, and neural-interface systems

Volume 03C of the Health Atlas Research Suite

Purpose

Connect neural structure, electrophysiology, circulation, networks, behavior and function to rigorous clinical reasoning and safe measurement, stimulation and neurotechnology engineering.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume is an evidence and engineering synthesis through 13 August 2026. It does not provide emergency protocols, drug dosing, surgical instructions, individual prognoses or permission to construct or apply a human-connected stimulator. Neurologic and psychiatric categories overlap; Volume 05 is the primary home for dimensional cognition, emotion and behavior, while this volume emphasizes nervous-system physiology, localization and neurotechnology.

How to use this volume

Nervous-system health is the capacity to sense, integrate, learn, decide, move, communicate, regulate internal organs and recover from perturbation. It cannot be summarized by a normal scan, a cognitive score or the absence of a focal lesion. A complete profile combines tempo and localization, electrophysiology, structure and perfusion, cognition and behavior, motor and sensory function, autonomic and sleep state, daily participation, exposures and longitudinal trajectory. Neural signals are especially vulnerable to artifact, state dependence, selection and overinterpretation. This volume therefore separates direct electrical measurement from hemodynamic and behavioral proxies, distinguishes diagnosis from monitoring, and treats neurotechnology as a coupled biological-software-governance system.

Questions this volume is designed to answer

1. How do electrical, chemical, vascular and network mechanisms create neurologic function?
2. How should localization and tempo guide testing?
3. What can EEG, EMG, imaging, CSF, wearables and implanted systems actually establish?
4. How do development, reserve, sleep, vascular health, immunity and environment shape trajectory?
5. How can neurodevices be validated without confusing correlation, decoding and causal control?
6. Which acute syndromes require time-critical escalation?
7. How should agency, privacy, continuity and disability rights be built into neurotechnology?

Scope

- Cellular excitability, synapses, glia, neural networks, perfusion, barriers and intracranial mechanics.
- Motor, sensory, autonomic, cognitive, language, sleep and plasticity systems.
- Acute vascular, electrical, traumatic, inflammatory, infectious, degenerative, neoplastic and functional disorders.
- Neurologic examination, electrophysiology, imaging, CSF, cognitive testing, wearables and invasive monitoring.
- Engineering and validation of EEG, EMG, movement, eye, stimulation, implant and brain-computer interface systems.
- Prevention, acute treatment, pharmacology, procedures, rehabilitation, assistive technology and supportive care.
- Ethics of neural data, agency, continuity and closed-loop intervention.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

Contents

Document status and intended use	i
How to use this volume	ii
1 Executive synthesis	1
1.1 What a complete profile requires	1
2 Historical development and changing models	3
2.0.1 Clinical descriptions of seizures, paralysis, head injury, memory change, pain and altered consciousness accumulated across medical traditions.	4
2.0.2 Anatomical dissection, microscopy, reflex experiments and electrical observations clarified brain, cord, nerves and muscle.	4
2.0.3 Clinicopathologic localization, neuron doctrine, histologic stains, aphasia syndromes and spinal pathways matured.	4
2.0.4 Electroencephalography, electrodiagnosis, cerebrospinal-fluid testing, angiography and neurosurgery expanded.	4
2.0.5 CT, MRI, Doppler, modern neuropathology, neurotransmitter pharmacology, intensive neurocritical care and rehabilitation developed.	5
2.0.6 Genetics, functional MRI, PET, high-density EEG, deep brain stimulation, thrombolysis/thrombectomy, biologics and molecular diagnosis.	5
2.0.7 Wearables, digital phenotyping, single-cell/spatial atlases, antisense/gene therapies, adaptive stimulation and AI-enabled imaging accelerated.	5
2.0.8 Neurologic health is increasingly represented through networks, development, reserve, function, environment and lived experience rather than lesion alone.	5
3 Normal foundations: structure, physiology, and regulation	7
3.0.1 Membrane excitability and action potentials	7
3.0.2 Synapses and neurotransmission	7
3.0.3 Neural coding and networks	8
3.0.4 Cerebral perfusion and metabolism	8
3.0.5 Barriers, cerebrospinal fluid and intracranial mechanics	8
3.0.6 Motor systems and neuromuscular execution	9
3.0.7 Somatosensation and pain	9
3.0.8 Autonomic and homeostatic regulation	9
3.0.9 Learning, memory and plasticity	10
3.0.10 Glia, myelin and immune-neural interfaces	10
3.0.11 Development, reserve and aging	11
3.1 Useful quantitative relationships	11

4	Determinants, heterogeneity, and causal architecture	12
4.0.1	Genetic architecture and family history	13
4.0.2	Development and life course	14
4.0.3	Nutrition and metabolism	14
4.0.4	Physical activity, conditioning, and sedentary behavior	14
4.0.5	Sleep, circadian timing, and recovery	15
4.0.6	Stress, emotion, and behavior	15
4.0.7	Social, economic, cultural, and care conditions	15
4.0.8	Environmental and occupational exposures	16
4.0.9	Health care, medication, and iatrogenic effects	16
5	Major disorders, failure modes, and clinically important phenotypes	17
5.0.1	Ischemic stroke and transient ischemic attack	17
5.0.2	Intracerebral and subarachnoid hemorrhage	17
5.0.3	Epilepsy and seizures	18
5.0.4	Migraine and primary headache disorders	18
5.0.5	Traumatic brain injury	19
5.0.6	Spinal cord injury and myelopathy	19
5.0.7	Multiple sclerosis and inflammatory demyelination	20
5.0.8	Peripheral neuropathy	20
5.0.9	Neuromuscular junction and muscle disease	21
5.0.10	Parkinson disease and parkinsonian syndromes	21
5.0.11	Alzheimer disease and related dementias	22
5.0.12	Motor neuron disease	22
5.0.13	Central nervous system infection and encephalitis	23
5.0.14	Brain and spinal tumors	23
5.0.15	Autonomic disorders and dysautonomia	24
5.0.16	Functional neurologic disorder	24
6	Engineering a nervous-system measurement stack	26
6.1	Noninvasive research platform	26
6.2	Do not collapse these distinctions	26
7	Neurotechnology governance and human agency	27
7.1	Governance requirements	27
8	Neurologic red flags and time-critical syndromes	28
8.1	Emergency escalation patterns	28
9	Assessment, clinical reasoning, and longitudinal profiling	29
9.0.1	1. Question and urgency	30

9.0.2	2. History and lived phenotype	30
9.0.3	3. Examination and basic measurements	30
9.0.4	4. Targeted laboratory, imaging, and physiology	30
9.0.5	5. Function and challenge testing	31
9.0.6	6. Longitudinal integration	31
10	Measurement and device landscape	32
10.0.1	Neurologic history and examination	35
10.0.2	Electroencephalography	35
10.0.3	Intracranial EEG and electrocorticography	36
10.0.4	Nerve conduction studies and electromyography	36
10.0.5	Evoked potentials	37
10.0.6	CT and CT angiography	37
10.0.7	MRI, diffusion, perfusion and spectroscopy	37
10.0.8	Functional MRI	38
10.0.9	PET and SPECT	38
10.0.10	Transcranial and carotid Doppler ultrasound	39
10.0.11	Lumbar puncture and CSF analysis	39
10.0.12	Intracranial pressure and multimodal neurocritical monitoring	39
10.0.13	Cognitive and neuropsychological assessment	40
10.0.14	Wearable movement, gait, tremor and sleep sensing	40
10.0.15	Eye tracking, pupillometry and oculography	41
10.0.16	Transcranial magnetic/electrical stimulation	41
10.0.17	Implanted neurostimulation and brain-computer interfaces	41
11	Engineering architectures, signal quality, and validation	43
11.0.1	Intended use and measurand	44
11.0.2	Sensor and transduction	44
11.0.3	Timing and synchronization	44
11.0.4	Analog, acquisition, and digital integrity	45
11.0.5	Calibration and traceability	45
11.0.6	Artifact and quality channels	45
11.0.7	Algorithms and uncertainty	46
11.0.8	Safety, usability, cybersecurity, and privacy	46
12	Prevention, treatment, rehabilitation, and system intervention	47
12.0.1	Vascular risk prevention	49
12.0.2	Time-critical reperfusion and hemorrhage care	49
12.0.3	Antiseizure medication and rescue therapy	49
12.0.4	Disease-modifying immunotherapy	50

12.0.5 Neurodegenerative symptomatic and targeted therapy	50
12.0.6 Neurosurgery and endovascular procedures	50
12.0.7 Neurostimulation and neuromodulation	51
12.0.8 Physical, occupational, speech and cognitive rehabilitation	51
12.0.9 Pain and symptom management	51
12.0.10 Sleep and circadian intervention	52
12.0.11 Nutrition, swallowing and respiratory support	52
12.0.12 Psychological, social and caregiver support	52
13 What is known, not known, and frequently overstated	53
13.0.1 Can digital motor and cognitive biomarkers replace clinic-based outcomes?	54
13.0.2 How should preclinical neurodegenerative biomarkers be used?	55
13.0.3 Can closed-loop stimulation discover individualized causal circuits?	55
13.0.4 What drives persistent symptoms after mild neurologic injury or infection?	55
13.0.5 How much risk is genetic versus environmental?	56
13.0.6 Can one composite score summarize health?	56
13.0.7 When is earlier detection beneficial?	56
13.0.8 Do digital biomarkers generalize?	57
13.0.9 What counts as meaningful change?	57
14 Research and development roadmap	58
14.0.1 0. Multimodal event truth set	59
14.0.2 0. Neurodevice continuity study	59
14.0.3 0. Recovery reserve model	60
14.0.4 1. Reference map	60
14.0.5 2. Causal determinants	60
14.0.6 3. Measurement validity	61
14.0.7 4. Clinical utility	61
14.0.8 5. Longitudinal multimorbidity	61
14.0.9 6. Implementation and governance	61
15 Cross-volume integration	63
A Minimum data schema for a longitudinal domain profile	64
B Glossary	65
Selected references and current standards	66

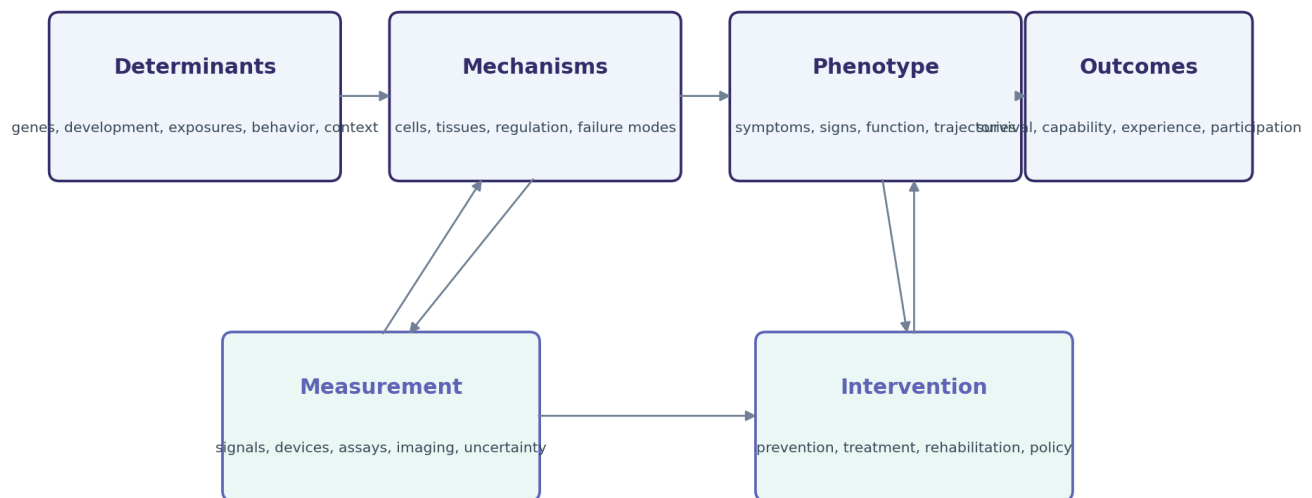
Executive synthesis

Nervous-system health is the capacity to sense, integrate, learn, decide, move, communicate, regulate internal organs and recover from perturbation. It cannot be summarized by a normal scan, a cognitive score or the absence of a focal lesion. A complete profile combines tempo and localization, electrophysiology, structure and perfusion, cognition and behavior, motor and sensory function, autonomic and sleep state, daily participation, exposures and longitudinal trajectory. Neural signals are especially vulnerable to artifact, state dependence, selection and overinterpretation. This volume therefore separates direct electrical measurement from hemodynamic and behavioral proxies, distinguishes diagnosis from monitoring, and treats neurotechnology as a coupled biological-software-governance system.

Central organizing principle

Connect neural structure, electrophysiology, circulation, networks, behavior and function to rigorous clinical reasoning and safe measurement, stimulation and neurotechnology engineering.

Nervous System Health and Neurotechnology



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.

- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Sudden focal deficit, thunderclap headache, prolonged seizure, rapidly declining consciousness, suspected meningitis/encephalitis, acute spinal compression, ascending weakness, or respiratory/bulbar failure requires urgent professional evaluation. Do not apply electrical or magnetic stimulation, implanted interfaces, or mains-referenced recording equipment to people outside approved systems and oversight.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Ancient to medieval	Clinical descriptions of seizures, paralysis, head injury, memory change, pain and altered consciousness accumulated across medical traditions.	Established reproducible neurologic phenomena and early localization from side-specific deficits.	Explanations mixed brain, heart, spirits and humors; anatomy and experimental access were limited.
1500s-1700s	Anatomical dissection, microscopy, reflex experiments and electrical observations clarified brain, cord, nerves and muscle.	Separated central and peripheral pathways and made neural signaling experimentally tractable.	Cell theory, synapses, ionic mechanisms and functional networks remained unknown.
1800s	Clinicopathologic localization, neuron doctrine, histologic stains, aphasia syndromes and spinal pathways matured.	Linked focal signs to lesions and established the neuron as a structural unit.	Static lesion maps underrepresented distributed networks, plasticity and context.
1900-1950	Electroencephalography, electrodiagnosis, cerebrospinal-fluid testing, angiography and neurosurgery expanded.	Made electrical dynamics, peripheral conduction and vascular anatomy measurable.	Signals were low-dimensional and invasive procedures carried substantial risk.
1950-1990	CT, MRI, Doppler, modern neuropathology, neurotransmitter pharmacology, intensive neurocritical care and rehabilitation developed.	Enabled rapid lesion detection, mechanistic treatment and survival after acute injury.	Imaging findings were often mistaken for complete explanation; disability outcomes remained secondary.
1990-2015	Genetics, functional MRI, PET, high-density EEG, deep brain stimulation, thrombolysis/thrombectomy, biologics and molecular diagnosis.	Reframed many disorders by pathways, circuits and time-critical intervention.	Biomarkers and genetic variants often had incomplete penetrance or limited clinical utility.
2015-2026	Wearables, digital phenotyping, single-cell/spatial atlases, antisense/gene therapies, adaptive stimulation and AI-enabled imaging accelerated.	Made continuous function, rare disease mechanisms and closed-loop intervention possible.	Dataset shift, opaque algorithms, long-term safety, cost and equity are unresolved.

Continued on next page

Period	Development	What it made possible	Residual limitation
Current systems view	Neurologic health is increasingly represented through networks, development, reserve, function, environment and lived experience rather than lesion alone.	Supports integration of cognition, movement, sensation, autonomic control, communication and participation.	Clinical systems still split neurology, psychiatry, rehabilitation, pain and social care despite shared mechanisms.

2.0.1 Clinical descriptions of seizures, paralysis, head injury, memory change, pain and altered consciousness accumulated across medical traditions.

Context. Neurology began as careful observation of function and failure.

Scientific consequence. Established reproducible neurologic phenomena and early localization from side-specific deficits.

What remained unresolved. Explanations mixed brain, heart, spirits and humors; anatomy and experimental access were limited.

Evidence anchors: [4,5]

2.0.2 Anatomical dissection, microscopy, reflex experiments and electrical observations clarified brain, cord, nerves and muscle.

Context. Structure increasingly constrained theory.

Scientific consequence. Separated central and peripheral pathways and made neural signaling experimentally tractable.

What remained unresolved. Cell theory, synapses, ionic mechanisms and functional networks remained unknown.

Evidence anchors: [3,4]

2.0.3 Clinicopathologic localization, neuron doctrine, histologic stains, aphasia syndromes and spinal pathways matured.

Context. The neurologic examination became a precision instrument.

Scientific consequence. Linked focal signs to lesions and established the neuron as a structural unit.

What remained unresolved. Static lesion maps underrepresented distributed networks, plasticity and context.

Evidence anchors: [4,5]

2.0.4 Electroencephalography, electrodiagnosis, cerebrospinal-fluid testing, angiography and neurosurgery expanded.

Context. Functional and structural measurements began to converge.

Scientific consequence. Made electrical dynamics, peripheral conduction and vascular anatomy measurable.

What remained unresolved. Signals were low-dimensional and invasive procedures carried substantial risk.

Evidence anchors: [6,8]

2.0.5 CT, MRI, Doppler, modern neuropathology, neurotransmitter pharmacology, intensive neurocritical care and rehabilitation developed.

Context. Neurology became increasingly technology- and specialty-divided.

Scientific consequence. Enabled rapid lesion detection, mechanistic treatment and survival after acute injury.

What remained unresolved. Imaging findings were often mistaken for complete explanation; disability outcomes remained secondary.

Evidence anchors: [7,16]

2.0.6 Genetics, functional MRI, PET, high-density EEG, deep brain stimulation, thrombolysis/thrombectomy, biologics and molecular diagnosis.

Context. Precision classification grew faster than causal treatment for many degenerative diseases.

Scientific consequence. Reframed many disorders by pathways, circuits and time-critical intervention.

What remained unresolved. Biomarkers and genetic variants often had incomplete penetrance or limited clinical utility.

Evidence anchors: [6,7,14]

2.0.7 Wearables, digital phenotyping, single-cell/spatial atlases, antisense/gene therapies, adaptive stimulation and AI-enabled imaging accelerated.

Context. Neurotechnology now links biology, software, behavior and governance.

Scientific consequence. Made continuous function, rare disease mechanisms and closed-loop intervention possible.

What remained unresolved. Dataset shift, opaque algorithms, long-term safety, cost and equity are unresolved.

Evidence anchors: [13,20,21]

2.0.8 Neurologic health is increasingly represented through networks, development, reserve, function, environment and lived experience rather than lesion alone.

Context. The atlas treats boundaries as useful administrative interfaces, not natural divisions.

Scientific consequence. Supports integration of cognition, movement, sensation, autonomic control, communication and participation.

What remained unresolved. Clinical systems still split neurology, psychiatry, rehabilitation, pain and social care despite shared mechanisms.

Evidence anchors: ^[10,11,17]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Membrane excitability and action potentials

Core principle. Ion gradients and voltage-gated channels permit rapid all-or-none signaling while graded potentials encode local input.

Control architecture. Na/K pumps, channel expression, myelination, extracellular ions, temperature and neuromodulators set thresholds and conduction.

Failure modes. Channelopathy, ischemia, toxins, demyelination, electrolyte disturbance or energy failure alters firing and propagation.

Observable consequences. Nerve conduction, EMG, EEG/ECOG, evoked responses, ion levels and electrophysiology.

Cross-system coupling. Metabolism, renal function, endocrine state and drugs directly alter excitability.

Evidence anchors: [3,4,25]

3.0.2 Synapses and neurotransmission

Core principle. Chemical and electrical synapses transform presynaptic activity into time- and state-dependent postsynaptic effects.

Control architecture. Release probability, receptor subtype, reuptake, degradation, plasticity and network inhibition/excitation regulate gain.

Failure modes. Receptor/channel disorders, autoimmune attack, toxins, degeneration, developmental imbalance or drug effects disrupt communication.

Observable consequences. EEG rhythms, evoked potentials, pharmacologic response, imaging ligands and cellular assays.

Cross-system coupling. Immune, endocrine, sleep, stress and pharmacologic systems modulate synapses.

Evidence anchors: [3,4,25]

3.0.3 Neural coding and networks

Core principle. Information is represented by firing rate, timing, synchrony, population state and recurrent network dynamics rather than one neuron per function.

Control architecture. Feedforward, feedback, inhibition, oscillation, neuromodulation and plasticity support flexible computation.

Failure modes. Disconnection, hypersynchrony, maladaptive plasticity or attractor instability can produce deficits or symptoms without gross lesions.

Observable consequences. EEG/MEG, intracranial recording, functional imaging, behavior and computational models.

Cross-system coupling. Cognition, emotion, movement and autonomic regulation share overlapping networks.

Evidence anchors: [4,10,46]

3.0.4 Cerebral perfusion and metabolism

Core principle. The brain requires continuous oxygen and substrate delivery with tight coupling between activity, blood flow and metabolism.

Control architecture. Autoregulation, neurovascular coupling, blood gases, systemic pressure, cardiac output and microcirculation maintain supply.

Failure modes. Ischemia, hypoxia, edema, hemorrhage, mitochondrial failure or impaired autoregulation causes rapid dysfunction and injury.

Observable consequences. Perfusion imaging, Doppler, oxygenation, pressure monitoring, metabolic imaging and neurologic state.

Cross-system coupling. Cardiovascular, respiratory, hematologic and metabolic systems determine reserve.

Evidence anchors: [2,4,7]

3.0.5 Barriers, cerebrospinal fluid and intracranial mechanics

Core principle. Blood-brain and blood-CSF barriers regulate exchange; CSF production, flow and absorption interact with fixed cranial volume.

Control architecture. Endothelium, glia, choroid plexus, venous drainage and pressure-volume compensation maintain environment.

Failure modes. Inflammation, hemorrhage, mass, hydrocephalus or venous obstruction disrupt barriers and raise pressure.

Observable consequences. CSF composition/pressure, imaging, optic nerve/retinal signs, invasive ICP and cerebral perfusion pressure.

Cross-system coupling. Immune, infectious, vascular and renal osmotic states interact.

Evidence anchors: [4,5,29]

3.0.6 Motor systems and neuromuscular execution

Core principle. Cortex, basal ganglia, cerebellum, brainstem, spinal circuits, peripheral nerves, neuromuscular junction and muscle transform goals into movement.

Control architecture. Sensory feedback, internal models, descending control, motor-unit recruitment and learning regulate force and coordination.

Failure modes. Weakness, spasticity, rigidity, ataxia, dystonia, tremor, apraxia or fatigue localize to different levels and mechanisms.

Observable consequences. Examination, kinematics, force, EMG/NCS, imaging, gait and task performance.

Cross-system coupling. Musculoskeletal mechanics, pain, cognition, motivation and cardiorespiratory reserve shape output.

Evidence anchors: [9,17]

3.0.7 Somatosensation and pain

Core principle. Peripheral receptors and central pathways encode touch, proprioception, temperature, itch and nociception; experience is modulated by context and prediction.

Control architecture. Descending inhibition/facilitation, attention, emotion, inflammation and learning regulate salience.

Failure modes. Nerve injury, central lesion, sensitization or disordered modulation creates loss, paresthesia or persistent pain.

Observable consequences. Bedside sensory testing, quantitative sensory testing, nerve studies, autonomic tests and patient-reported impact.

Cross-system coupling. Immune, skin, musculoskeletal, cognitive and psychosocial systems are inseparable.

Evidence anchors: [10,17,27]

3.0.8 Autonomic and homeostatic regulation

Core principle. Central and peripheral autonomic networks regulate heart rate, pressure, respiration, digestion, thermoregulation, bladder, sexual function and energy balance.

Control architecture. Baroreflexes, chemoreflexes, hypothalamus, brainstem, spinal and peripheral ganglia integrate internal state and behavior.

Failure modes. Neurodegeneration, neuropathy, injury, autoimmune disease or drugs produce orthostatic, visceral and thermoregulatory failure.

Observable consequences. Orthostatic vitals, beat-to-beat pressure, HRV with caution, sudomotor tests, pupillometry and organ-specific studies.

Cross-system coupling. Every organ-system volume intersects autonomic physiology.

Evidence anchors: [2,4]

3.0.9 Learning, memory and plasticity

Core principle. Experience changes synaptic strength, network structure and behavior across development, skill learning and recovery.

Control architecture. Attention, sleep, neuromodulators, repetition, reward, error signals and environmental enrichment shape plasticity.

Failure modes. Degeneration, injury, stress, sleep disruption or maladaptive reinforcement impairs learning or creates persistent symptoms.

Observable consequences. Neuropsychological tasks, longitudinal function, imaging, electrophysiology and ecological performance.

Cross-system coupling. Mental health, education, rehabilitation, social context and aging determine expression.

Evidence anchors: [10,15,16]

3.0.10 Glia, myelin and immune-neural interfaces

Core principle. Astrocytes, oligodendrocytes, microglia and peripheral glia support metabolism, myelination, repair and immune surveillance.

Control architecture. Cell-cell signaling, trophic support, myelin turnover, barrier integrity and inflammatory resolution regulate networks.

Failure modes. Demyelination, gliosis, neuroinflammation, infection or defective support slows conduction and promotes degeneration.

Observable consequences. MRI, CSF, evoked potentials, pathology, molecular imaging and fluid biomarkers.

Cross-system coupling. Immunity, infection, aging and metabolism converge on glial biology.

Evidence anchors: [13,15,29]

3.0.11 Development, reserve and aging

Core principle. Neurodevelopment builds circuits through proliferation, migration, pruning and experience; reserve and compensation buffer later pathology.

Control architecture. Genes, prenatal environment, attachment, education, activity, sleep, vascular health and plasticity influence trajectory.

Failure modes. Developmental disruption, cumulative injury, vascular burden, proteinopathy and cellular aging reduce reserve.

Observable consequences. Milestones, cognition, education/occupation history, imaging, function, sensory status and longitudinal change.

Cross-system coupling. Life course, maternal health, psychosocial conditions and all organ systems contribute.

Evidence anchors: [15,36,37]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Nernst potential	$E_i = \frac{RT}{z_i F} \ln \frac{[i]_o}{[i]_i}$	Equilibrium voltage for one ion reflects concentration ratio, charge and temperature.	Real membranes contain multiple conductances and active transport; concentrations are compartment- and state-dependent.
Membrane time constant	$\tau_m = R_m C_m$	Sets the speed with which membrane voltage responds to current.	Neurons are spatial cables; dendritic geometry and active channels violate a single-compartment model.
Cable length constant	$\lambda \propto \sqrt{r_m / r_i}$	Relates passive spread to membrane and axial resistance.	Myelin, branching, active conductances and nonuniform geometry require richer models.
Cerebral perfusion pressure	$CPP \approx MAP - ICP$	Approximate pressure gradient driving cerebral perfusion.	Venous pressure, autoregulation, regional heterogeneity and measurement site matter; a target is not universal.
Information	$I(x) = -\log_2 p(x)$	A less probable event carries more Shannon information in a defined coding model.	Information-theoretic value is not equivalent to biological meaning or clinical utility.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within nervous system health and neurotechnology.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Continued on next page

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within nervous system health and neurotechnology.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: [3,14]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

Evidence anchors: [36,37]

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[28]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[40]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[26]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the neurologic syndrome, lesion, network disorder, or functional phenotype, but cross-system processes and functional effects are retained.

5.0.1 Ischemic stroke and transient ischemic attack

Definition and phenotype. Acute focal neurologic dysfunction caused by cerebral, retinal or spinal ischemia; transient symptoms may occur without completed infarction.

Pathophysiology. Arterial occlusion or critical hypoperfusion causes energy failure, excitotoxicity, ionic edema, inflammation and infarction; collateral flow and time determine salvage.

Determinants and modifiers. Atherosclerosis, atrial fibrillation, small-vessel disease, dissection, thrombophilia, hypertension, diabetes, tobacco, age and inequitable prevention/response.

Assessment. Immediate time-of-onset and neurologic examination; glucose; CT/MRI and vascular imaging; cardiac/vascular etiologic workup; disability and swallowing assessment.

Intervention logic. Time-critical reperfusion for eligible patients, antithrombotic and cause-specific prevention, risk-factor management, rehabilitation and secondary prevention.

Known limits and open questions. Imaging-defined tissue windows improve selection but do not eliminate uncertainty; cryptogenic mechanisms and recovery prediction remain imperfect.

Suite links. 03A cardiovascular; 03G thrombosis; 10 rehabilitation; 12 emergency systems.

Evidence anchors: [2,7,33]

5.0.2 Intracerebral and subarachnoid hemorrhage

Definition and phenotype. Bleeding into brain, ventricles or subarachnoid space causing direct injury, pressure effects and secondary ischemia.

Pathophysiology. Vessel rupture produces mass, toxicity, edema, hydrocephalus, vasospasm or rebleeding.

Determinants and modifiers. Hypertension, aneurysm, vascular malformation, amyloid angiopathy, anticoagulation, drugs, trauma and coagulopathy.

Assessment. Emergency CT/angiography, neurologic and hemodynamic assessment, coagulation and etiologic evaluation, ICP/hydrocephalus monitoring when indicated.

Intervention logic. Reverse coagulopathy, control pressure within context, neurosurgical/endovascular management, prevent complications and rehabilitate.

Known limits and open questions. Optimal intervention varies by location, timing and physiology; prognostic scores should not become self-fulfilling withdrawal rules.

Suite links. 03A vascular; 03G hemostasis; 15 ethics; 12 neurocritical care.

Evidence anchors: [7,26,33]

5.0.3 Epilepsy and seizures

Definition and phenotype. A seizure is transient abnormal synchronized brain activity; epilepsy is an enduring predisposition to recurrent unprovoked seizures or defined syndromes.

Pathophysiology. Network excitability and synchronization arise from genetic, structural, metabolic, immune, infectious or unknown mechanisms.

Determinants and modifiers. Development, brain injury, stroke, tumor, infection, sleep loss, drugs, metabolic disturbance and genotype modify risk.

Assessment. Event history/video and witness account, EEG, MRI, laboratory or genetic/immune testing as indicated; distinguish mimics and classify syndrome.

Intervention logic. Acute rescue when prolonged; antiseizure medication by seizure/syndrome and person; surgery, stimulation or diet for selected drug-resistant disease; safety and psychosocial support.

Known limits and open questions. Interictal EEG can be normal or nonspecific; wearables detect only selected motor/autonomic patterns; treatment withdrawal prediction is uncertain.

Suite links. 05 mental health; 07 development; 08 infection/immunity; 14 pharmacology/neurodevices.

Evidence anchors: [6,8,22]

5.0.4 Migraine and primary headache disorders

Definition and phenotype. Recurrent headache and associated sensory, autonomic or neurologic symptoms without a structural lesion explaining the attacks.

Pathophysiology. Trigeminovascular activation, brainstem/hypothalamic networks, cortical excitability and genetic susceptibility interact with state and triggers.

Determinants and modifiers. Family history, hormones, sleep, stress, sensory load, medication overuse, diet patterns and comorbidity.

Assessment. Phenotype and red-flag history; neurologic examination; imaging only when indicated; headache diary and disability assessment.

Intervention logic. Education and trigger regularity, acute therapy, preventive drugs or neuromodulation by burden and preference, avoidance of medication overuse.

Known limits and open questions. Triggers are not universal causes; biomarkers remain limited; placebo/context effects and fluctuating baseline complicate trials.

Suite links. 05 emotion/behavior; 03M reproductive; 14 pharmacology; 10 function.

Evidence anchors: [6,27]

5.0.5 Traumatic brain injury

Definition and phenotype. Brain dysfunction or structural injury caused by external force, ranging from concussion to severe diffuse and focal damage.

Pathophysiology. Mechanical deformation, axonal injury, hemorrhage, metabolic crisis, inflammation, edema and secondary hypoxia/ischemia.

Determinants and modifiers. Impact mechanics, protective systems, age, prior injury, alcohol/drugs, anticoagulation, military/work/sport exposure and access to care.

Assessment. Acute severity and red flags, CT when indicated, serial examination, symptom/cognitive/vestibular assessment and functional trajectory.

Intervention logic. Prevent secondary injury, symptom-targeted graded return, rehabilitation, mental-health and sleep care; surgery/critical care for severe lesions.

Known limits and open questions. Concussion lacks a single diagnostic biomarker; persistent symptoms are heterogeneous; premature return and overrestriction both cause harm.

Suite links. 04 trauma; 05 cognition; 09 occupation; 10 rehabilitation.

Evidence anchors: [16,26]

5.0.6 Spinal cord injury and myelopathy

Definition and phenotype. Damage or dysfunction of spinal cord from trauma, compression, vascular, inflammatory, degenerative or metabolic causes.

Pathophysiology. Tract and segment injury disrupts motor, sensory, autonomic and respiratory pathways; secondary edema and ischemia can extend damage.

Determinants and modifiers. Trauma, degenerative stenosis, tumor, infection, autoimmune disease, vascular lesions and nutritional deficiency.

Assessment. Level/localization examination, urgent MRI for compression, CT for bone, laboratory/CSF as appropriate, bladder/bowel/respiratory and function assessment.

Intervention logic. Emergency stabilization and decompression when indicated, cause-specific therapy, prevention of pressure/thrombotic/autonomic complications, intensive rehabilitation and assistive technology.

Known limits and open questions. Recovery varies by spared pathways and complications; imaging severity incompletely predicts function; neuroregeneration claims require rigorous trials.

Suite links. 03I musculoskeletal; 03F urinary; 10 rehabilitation/assistive tech; 15 disability rights.

Evidence anchors: [6,16,19]

5.0.7 Multiple sclerosis and inflammatory demyelination

Definition and phenotype. Immune-mediated central nervous-system demyelination and neuroaxonal injury with relapsing, progressive or monophasic phenotypes.

Pathophysiology. Autoreactive immune activity, barrier disruption, demyelination, gliosis and axonal loss interact with repair failure.

Determinants and modifiers. Genetic susceptibility, latitude/vitamin D correlates, smoking, prior infection, obesity and sex-related factors modify risk; causal attribution is complex.

Assessment. Clinical dissemination in time/space, MRI, CSF and exclusion of mimics; disability, cognition, vision and function over time.

Intervention logic. Relapse treatment, disease-modifying therapy by activity/risk, symptom treatment, rehabilitation, vaccination and monitoring for adverse effects.

Known limits and open questions. Biomarkers of progression and treatment de-escalation are incomplete; MRI activity is not identical to lived disability.

Suite links. 03H immunity; 03K vision; 10 rehabilitation; 14 immunotherapy.

Evidence anchors: [6,14,34]

5.0.8 Peripheral neuropathy

Definition and phenotype. Damage to peripheral sensory, motor or autonomic nerves in length-dependent, focal, multifocal or generalized patterns.

Pathophysiology. Axonal degeneration, demyelination, ischemia, metabolic toxicity, immune attack, compression, infection or inherited defects.

Determinants and modifiers. Diabetes, alcohol/toxins/drugs, nutritional deficiency, kidney disease, autoimmune disease, infection, genetics and mechanical entrapment.

Assessment. Pattern-focused examination, nerve conduction/EMG, laboratory cause evaluation, autonomic or small-fiber tests, genetic/biopsy studies selectively.

Intervention logic. Treat cause, pain and autonomic symptoms; protect insensate tissue; rehabilitation and orthoses; immune therapy for defined inflammatory neuropathies.

Known limits and open questions. Small-fiber neuropathy lacks one universal reference; causal attribution in common metabolic/toxic settings can be probabilistic.

Suite links. 03D diabetes/metabolism; 03F renal; 03I function; 03H immune.

Evidence anchors: [6,27,35]

5.0.9 Neuromuscular junction and muscle disease

Definition and phenotype. Disorders of motor unit transmission or muscle cause weakness, fatigue, respiratory or bulbar failure, pain or metabolic crisis.

Pathophysiology. Autoantibodies, genetic defects, channel dysfunction, inflammation, degeneration, mitochondrial or metabolic defects impair force production.

Determinants and modifiers. Genetics, autoimmunity, drugs/toxins, endocrine/electrolyte states, infection and age.

Assessment. Distribution and fatigability; CK and targeted laboratories; EMG/NCS and repetitive stimulation; antibodies, imaging, genetics or biopsy as indicated; respiratory/swallow testing.

Intervention logic. Cause-specific immune, molecular or metabolic treatment; avoid harmful drugs; respiratory/nutritional support, rehabilitation and genetic counseling.

Known limits and open questions. Variant interpretation and treatment durability remain difficult; functional benefit may not track molecular correction.

Suite links. 03I muscle; 03B respiratory pump; 02 genomics; 14 advanced therapy.

Evidence anchors: [6,14,16]

5.0.10 Parkinson disease and parkinsonian syndromes

Definition and phenotype. Progressive disorders characterized by bradykinesia with tremor/rigidity and variable cognitive, autonomic, sleep and gait features.

Pathophysiology. Dopaminergic degeneration and distributed proteinopathy/network dysfunction; atypical syndromes involve additional systems.

Determinants and modifiers. Age, genetics, toxins/exposures and stochastic biology; many associations are noncausal or modest.

Assessment. Clinical examination and longitudinal response; imaging mainly to exclude alternatives or support selected uncertainty; cognition, mood, autonomic and function assessment.

Intervention logic. Dopaminergic therapy, exercise/rehabilitation, nonmotor treatment, deep brain stimulation or infusion therapies for selected people, safety planning.

Known limits and open questions. No routine test proves idiopathic disease; progression and treatment response vary; digital motor biomarkers need utility validation.

Suite links. 05 cognition/mood; 07 aging; 09 environment; 10 rehabilitation.

Evidence anchors: [6,9,22]

5.0.11 Alzheimer disease and related dementias

Definition and phenotype. Acquired decline in one or more cognitive domains sufficient to affect independence, due to neurodegenerative, vascular, mixed or other disease.

Pathophysiology. Protein aggregation, synaptic/neuronal loss, inflammation, vascular injury and network disconnection evolve over years.

Determinants and modifiers. Age, genetic variants, vascular/metabolic risk, education/reserve, hearing loss, sleep, social context and traumatic injury.

Assessment. History from patient and informant, cognitive and functional testing, medication/mood/sensory review, laboratory and imaging; biomarkers when they change counseling or therapy.

Intervention logic. Risk-factor and safety management, cognitive/functional support, caregiver and social services, symptomatic drugs, and disease-modifying therapy in narrowly eligible stages with monitoring.

Known limits and open questions. Biomarker positivity is not equivalent to symptomatic inevitability; effect size, imaging adverse events, access and long-term meaningful benefit remain debated.

Suite links. 05 cognition; 07 aging; 03A vascular; 06 social/caregiver; 14 therapeutics.

Evidence anchors: [6,15,18]

5.0.12 Motor neuron disease

Definition and phenotype. Progressive degeneration of upper and/or lower motor neurons causing weakness, bulbar and respiratory failure with variable cognitive/behavioral involvement.

Pathophysiology. Protein homeostasis, RNA processing, excitotoxic, mitochondrial, axonal and glial mechanisms vary by genetic and sporadic subtype.

Determinants and modifiers. Age, genetics and incompletely understood exposures; most cases lack a single modifiable cause.

Assessment. Clinical progression, EMG, exclusion of mimics, genetics when appropriate, respiratory/nutrition/communication and cognition assessment.

Intervention logic. Multidisciplinary care, disease-modifying drugs with modest/subtype-specific effects, noninvasive ventilation, nutrition, communication technology, symptom and palliative care.

Known limits and open questions. Heterogeneous molecular pathways and late diagnosis impede trials; surrogate biomarkers need validation against function and survival.

Suite links. 03B respiratory; 10 assistive tech; 06 family/caregiver; 14 therapeutics.

Evidence anchors: [6,18,42]

5.0.13 Central nervous system infection and encephalitis

Definition and phenotype. Infectious or immune-mediated inflammation of meninges, brain, cord or associated structures.

Pathophysiology. Pathogen invasion or autoimmune response causes barrier disruption, edema, seizures, vascular injury and neuronal dysfunction.

Determinants and modifiers. Pathogen exposure, vaccination, immune status, vectors, geography, procedures and delays in care.

Assessment. Emergency syndrome recognition; blood and CSF studies when safe, neuroimaging, EEG and targeted microbiologic/immune testing; monitor pressure and systemic organ failure.

Intervention logic. Empiric time-critical antimicrobial therapy when indicated, immune therapy for defined autoimmune disease, seizure/pressure support and rehabilitation.

Known limits and open questions. Negative tests do not exclude disease; antibodies may be nonspecific; post-infectious symptoms and relapse mechanisms vary.

Suite links. 08 infection/immunity; 12 acute care; 03H autoimmunity.

Evidence anchors: [6,29,30]

5.0.14 Brain and spinal tumors

Definition and phenotype. Primary or metastatic neoplasms can cause seizures, focal deficits, cognitive change, pressure effects and systemic complications.

Pathophysiology. Clonal growth, invasion, edema, hemorrhage, treatment injury and network disruption; biology varies widely by molecular class.

Determinants and modifiers. Genetic syndromes, prior radiation, immune state and systemic cancer; many cases have no identifiable preventable cause.

Assessment. Contrast MRI/CT, pathology and molecular classification, systemic staging, neurologic/functional baseline and treatment-risk mapping.

Intervention logic. Surgery, radiation and systemic/targeted/immune therapy by type and goals; seizure, edema, rehabilitation and palliative support.

Known limits and open questions. Imaging response may reflect treatment effect or pseudoprogression; tissue sampling and molecular heterogeneity limit certainty.

Suite links. 04 cancer; 14 therapeutics; 10 rehabilitation/palliative.

Evidence anchors: [14,31,32]

5.0.15 Autonomic disorders and dysautonomia

Definition and phenotype. Failure or dysregulation of autonomic control produces orthostatic intolerance, syncope, heart-rate, sweating, gastrointestinal, bladder, sexual or thermoregulatory symptoms.

Pathophysiology. Central or peripheral autonomic injury, neuropathy, immune disease, hypovolemia, hyperadrenergic state, medication effects or deconditioning.

Determinants and modifiers. Diabetes, neurodegeneration, infection/post-acute states, autoimmune disease, connective tissue disorders, drugs, bed rest and uncertain factors.

Assessment. Orthostatic history and standardized vitals; ECG/cardiac causes; autonomic reflex, tilt, sudomotor or catecholamine tests selectively; function and volume status.

Intervention logic. Cause treatment, education, fluids/salt or compression when appropriate, physical reconditioning and selected drugs; avoid aggravating agents.

Known limits and open questions. Syndrome labels can combine mechanisms; heart-rate thresholds alone do not establish cause; uncontrolled studies dominate some treatment claims.

Suite links. 03A cardiovascular; 03D metabolic; 03E/03F visceral systems; 05 anxiety overlap.

Evidence anchors: [2,6,45]

5.0.16 Functional neurologic disorder

Definition and phenotype. Genuine motor, sensory or seizure-like symptoms arising from altered nervous-system functioning rather than a lesion sufficient to explain the phenotype.

Pathophysiology. Abnormal attention, prediction, agency, salience, learning and motor control interact with biological and psychosocial factors.

Determinants and modifiers. Predisposing and precipitating factors vary; trauma or stress may be present but is neither necessary nor sufficient.

Assessment. Positive internally inconsistent or incongruent examination signs, event semiology and appropriate exclusion of important alternatives; formulation and function assessment.

Intervention logic. Clear nonstigmatizing explanation, physiotherapy/occupational/speech approaches by symptom, psychological therapy when indicated, and treatment of comorbidity.

Known limits and open questions. Mechanistic biomarkers are not diagnostic; dual diagnosis with neurologic disease is possible; care fragmentation worsens outcomes.

Suite links. 05 brain/behavior; 06 psychosocial; 10 rehabilitation; 15 stigma/equity.

Evidence anchors: ^[10,12,16]

Engineering a nervous-system measurement stack

Neural systems are high-dimensional, state-dependent and only partially observable. A practical platform should combine a small number of high-integrity signals with synchronized task, context and functional outcomes rather than assuming more electrodes or sensors create more truth.

6.1 Noninvasive research platform

- High-quality synchronized EEG with ECG/EOG/EMG and impedance/contact channels.
- Inertial and pressure sensing for gait, tremor, posture and task context.
- Eye tracking or electro-oculography with calibrated visual stimuli.
- Audio/video only with explicit consent, privacy controls and event annotation.
- Validated cognitive tasks with alternate forms and sensory/motor accommodations.
- Autonomic channels such as beat-to-beat pressure or skin conductance only for narrow claims.
- Common clock, raw export, event triggers, firmware/model provenance and a manual annotation interface.

Selected evidence anchors: [20,22,45]

6.2 Do not collapse these distinctions

Observation	Potential interpretations	Required discriminators
Reduced movement	Weakness, bradykinesia, pain, fatigue, depression, environment or device nonwear.	Exam, task context, force, symptoms, medication state and quality channels.
Abnormal EEG transient	Epileptiform activity, artifact, benign variant or nonspecific encephalopathy.	Expert review, montage/context, video, repetition and clinical event.
Cognitive score decline	Disease, sleep/mood, sensory loss, practice/test form, medication or acute illness.	Reliable-change methods, collateral function, state measures and longitudinal confirmation.
Orthostatic tachycardia	Volume loss, medication, deconditioning, autonomic neuropathy, hyperadrenergic state or anxiety.	Pressure, symptoms, standardized posture, labs/context and repeated testing.

Selected evidence anchors: [6,10,45]

Neurotechnology governance and human agency

Neural data can reveal health, behavior and identity-relevant patterns, while implanted systems can influence movement, mood or communication. Safety therefore includes agency, autonomy, privacy, device continuity and the right to meaningful human oversight.

7.1 Governance requirements

- Separate therapeutic control, research recording and secondary data use.
- Specify who can change stimulation, decode signals, update models or disable the device.
- Plan for explant, battery depletion, company failure, software obsolescence and loss of clinical support.
- Prevent coercive workplace, insurance, forensic or consumer reuse of neural data.
- Report uncertainty and maintain a nonalgorithmic communication or control fallback where possible.
- Include people with disability and device users in design and oversight.

Selected evidence anchors: [19,20,23]

Neurologic red flags and time-critical syndromes

Remote or experimental monitoring must never substitute for urgent assessment of sudden or rapidly progressive neurologic dysfunction.

8.1 Emergency escalation patterns

- Sudden face/arm/leg weakness, speech or vision change, severe imbalance or other focal deficit.
- First seizure, status epilepticus, repeated seizures without recovery or injury during seizure.
- Thunderclap headache, meningismus, fever with altered mental status or rapidly progressive encephalopathy.
- New spinal cord compression pattern, urinary retention with saddle symptoms, or rapidly ascending weakness.
- Head injury with worsening headache, vomiting, anticoagulation, seizure, focal deficit or declining consciousness.
- Acute respiratory/bulbar weakness, autonomic crisis or severe medication/toxin effect.

Selected evidence anchors: [6,7,26]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current nervous system health and neurotechnology risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

9.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current nervous system health and neurotechnology risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

Evidence anchors: ^[44]

9.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[17,41]

9.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[4,5]

9.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

Evidence anchors: ^[44]

9.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

Evidence anchors: ^[17]

9.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

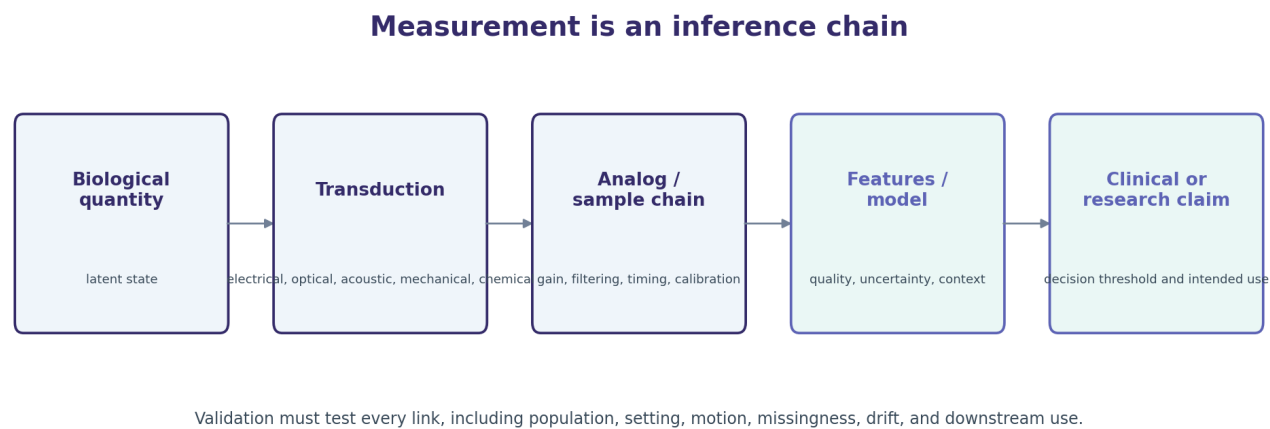


Figure 10.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Neurologic history and examination	Noninvasive	Structured observation and tasks probe mental status, cranial nerves, motor, reflex, sensory, coordination, gait and autonomic function.	Localization, tempo, severity, safety, differential and functional baseline.	High information density, immediate and repeatedly usable.	Examiner skill, fatigue, language, pain, attention, culture and interrater variability; normal findings do not exclude episodic disease.
Electroencephalography	Noninvasive electrical	Scalp electrodes measure summed postsynaptic field potentials; ambulatory/video systems add duration and behavior.	Rhythms, epileptiform activity, seizures, encephalopathy and event correlation.	Millisecond temporal resolution and direct electrical physiology.	Poor spatial specificity, skull filtering, artifact, sampling limitations and nonspecific abnormalities.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Intracranial EEG and electro-corticography	Invasive electrical	Depth or surface electrodes record local field potentials and stimulation responses.	Seizure networks, cortical mapping, high-resolution dynamics and control signals.	High signal quality and spatial-temporal specificity near contacts.	Hemorrhage, infection, limited coverage, sampling bias and ethical challenges.
Nerve conduction studies and electromyography	Minimally invasive electrical	Surface stimulation/recording and needle electrodes measure nerve and motor-unit physiology.	Axonal vs demyelinating patterns, localization, neuromuscular transmission and muscle activation.	Direct functional localization in peripheral system.	Discomfort, temperature, anatomy and operator dependence; small-fiber disorders may be missed.
Evoked potentials	Noninvasive electrical	Time-locked EEG or peripheral responses to visual, auditory, somatosensory or motor stimulation.	Conduction delay, pathway integrity and intraoperative monitoring.	Can reveal subclinical pathway dysfunction.	Small signals, averaging/artifact, nonspecific delay and limited localization.
CT and CT angiography	Noninvasive/minimally invasive; ionizing	X-ray tomography with or without iodinated contrast.	Hemorrhage, fracture, mass effect, hydrocephalus, large-vessel anatomy and perfusion proxies.	Fast and widely available for emergencies.	Radiation, contrast risk, limited soft-tissue detail and protocol dependence.
MRI, diffusion, perfusion and spectroscopy	Noninvasive/minimally invasive with contrast	Magnetic resonance of tissue properties, diffusion, flow/perfusion and metabolites.	Lesion anatomy, ischemia, demyelination, tumor, connectivity and selected chemistry.	Excellent soft-tissue contrast and versatile sequences without ionizing radiation.	Time, motion, implants, claustrophobia, contrast considerations and sequence/scanner dependence; incidental findings.
Functional MRI	Noninvasive	Blood-oxygen-level dependent or perfusion signal as an indirect proxy for neural activity.	Task/rest network maps, presurgical language/motor localization and research phenotypes.	Whole-brain spatial mapping.	Hemodynamic rather than direct neural signal; motion, physiology, analytic flexibility and poor individual transport.
PET and SPECT	Minimally invasive ionizing molecular imaging	Radiotracers measure metabolism, perfusion, transporters, receptors or pathologic proteins.	Tumor, seizure focus, neurodegeneration biomarkers and selected movement/cognitive differentials.	Molecular specificity unavailable to structural imaging.	Radiation, cost, tracer specificity, partial-volume effects and biomarker-disease ambiguity.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Transcranial and carotid Doppler ultrasound	Noninvasive acoustic	Ultrasound frequency shifts estimate blood velocity in accessible vessels.	Flow velocity, stenosis proxies, emboli, vasospasm and dynamic autoregulation.	Portable, repeated and high temporal resolution.	Operator/window dependence; velocity is not flow without geometry; extracranial/intracranial coverage limits.
Lumbar puncture and CSF analysis	Invasive	Pressure measurement and biochemical, cellular, microbiologic, immunologic or molecular assays of CSF.	Infection, inflammation, hemorrhage, pressure and selected neurodegenerative biomarkers.	Samples a compartment close to CNS biology.	Headache, bleeding/infection risk, pressure-position effects, contamination, preanalytic variability and nonspecific biomarkers.
Intracranial pressure and multimodal neurocritical monitoring	Invasive	Ventricular/parenchymal pressure sensors plus optional oxygen, flow, EEG and metabolism probes.	ICP, CPP and local physiologic trends.	Continuous data in severe injury and ability to drain CSF with ventricular systems.	Infection/bleeding, calibration drift, local sampling and uncertain treatment thresholds across phenotypes.
Cognitive and neuropsychological assessment	Noninvasive behavioral	Standardized tasks measure attention, memory, language, executive, visuospatial and processing domains.	Domain profile, validity, functional implications and longitudinal change.	Directly assesses performance relevant to cognition.	Education, language, culture, sensory/motor limitations, effort, practice effects and ecological gap.
Wearable movement, gait, tremor and sleep sensing	Noninvasive digital	Inertial, pressure, optical, location and interaction sensors quantify movement and context.	Activity, gait features, tremor, falls, medication cycles and selected sleep proxies.	Long-duration real-world sampling.	Device placement, adherence, algorithm opacity, environmental confounding and inability to infer cause.
Eye tracking, pupillometry and oculography	Noninvasive optical/electrical	Camera or electro-oculography captures gaze, saccades, nystagmus and pupil dynamics.	Cranial/brainstem/cerebellar function, attention, autonomic and fatigue proxies.	Sensitive high-frequency motor output and bedside portability.	Lighting, glasses, eyelid, pigmentation/iris characteristics, calibration and drug effects.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Transcranial magnetic/electrical stimulation	Noninvasive stimulation with physiological readout	Magnetic or electrical fields perturb cortical excitability; EMG/EEG or behavior measures response.	Motor thresholds, inhibition/facilitation, causal circuit tests and treatment delivery.	Can probe and modulate circuits without surgery.	Coarse fields, state dependence, discomfort, seizure risk and variable individual targeting.
Implanted neurostimulation and brain-computer interfaces	Invasive electrical	Deep, cortical, spinal or peripheral electrodes record and/or stimulate neural tissue.	Symptom control, closed-loop biomarkers, communication or motor control.	Direct access and potential closed-loop adaptation.	Surgery, infection, lead failure, tissue response, battery/cybersecurity, long-term data rights and dependence.

10.0.1 Neurologic history and examination

Measured quantity and transducer. Structured observation and tasks probe mental status, cranial nerves, motor, reflex, sensory, coordination, gait and autonomic function.

Typical outputs. Localization, tempo, severity, safety, differential and functional baseline.

Strengths. High information density, immediate and repeatedly usable.

Limitations and artifacts. Examiner skill, fatigue, language, pain, attention, culture and interrater variability; normal findings do not exclude episodic disease.

Reference and validation strategy. Standardized maneuvers/scales, observer training, blinded reliability and outcome correlation.

Clinical/research maturity. Mature foundational method.

Evidence anchors: [6,17]

10.0.2 Electroencephalography

Measured quantity and transducer. Scalp electrodes measure summed postsynaptic field potentials; ambulatory/video systems add duration and behavior.

Typical outputs. Rhythms, epileptiform activity, seizures, encephalopathy and event correlation.

Strengths. Millisecond temporal resolution and direct electrical physiology.

Limitations and artifacts. Poor spatial specificity, skull filtering, artifact, sampling limitations and non-specific abnormalities.

Reference and validation strategy. Electrode/impedance standards, amplifier testing, event annotations, expert agreement and clinical reference.

Clinical/research maturity. Mature; AI interpretation remains claim-specific.

Evidence anchors: [8,21,24]

10.0.3 Intracranial EEG and electrocorticography

Measured quantity and transducer. Depth or surface electrodes record local field potentials and stimulation responses.

Typical outputs. Seizure networks, cortical mapping, high-resolution dynamics and control signals.

Strengths. High signal quality and spatial-temporal specificity near contacts.

Limitations and artifacts. Hemorrhage, infection, limited coverage, sampling bias and ethical challenges.

Reference and validation strategy. Hardware safety, localization, independent clinical adjudication and longitudinal outcome.

Clinical/research maturity. Mature specialized clinical method; research BCI frontier.

Evidence anchors: [8,20,24]

10.0.4 Nerve conduction studies and electromyography

Measured quantity and transducer. Surface stimulation/recording and needle electrodes measure nerve and motor-unit physiology.

Typical outputs. Axonal vs demyelinating patterns, localization, neuromuscular transmission and muscle activation.

Strengths. Direct functional localization in peripheral system.

Limitations and artifacts. Discomfort, temperature, anatomy and operator dependence; small-fiber disorders may be missed.

Reference and validation strategy. Electrical calibration, normative data, temperature control, repeatability and clinical correlation.

Clinical/research maturity. Mature specialist method.

Evidence anchors: [6,24]

10.0.5 Evoked potentials

Measured quantity and transducer. Time-locked EEG or peripheral responses to visual, auditory, somatosensory or motor stimulation.

Typical outputs. Conduction delay, pathway integrity and intraoperative monitoring.

Strengths. Can reveal subclinical pathway dysfunction.

Limitations and artifacts. Small signals, averaging/artifact, nonspecific delay and limited localization.

Reference and validation strategy. Stimulus timing, amplifier integrity, normative distributions and outcome-linked thresholds.

Clinical/research maturity. Established adjunct.

Evidence anchors: [6]

10.0.6 CT and CT angiography

Measured quantity and transducer. X-ray tomography with or without iodinated contrast.

Typical outputs. Hemorrhage, fracture, mass effect, hydrocephalus, large-vessel anatomy and perfusion proxies.

Strengths. Fast and widely available for emergencies.

Limitations and artifacts. Radiation, contrast risk, limited soft-tissue detail and protocol dependence.

Reference and validation strategy. Dose/phantom QA, timing, reconstruction version, blinded comparison and workflow outcomes.

Clinical/research maturity. Mature.

Evidence anchors: [7,26]

10.0.7 MRI, diffusion, perfusion and spectroscopy

Measured quantity and transducer. Magnetic resonance of tissue properties, diffusion, flow/perfusion and metabolites.

Typical outputs. Lesion anatomy, ischemia, demyelination, tumor, connectivity and selected chemistry.

Strengths. Excellent soft-tissue contrast and versatile sequences without ionizing radiation.

Limitations and artifacts. Time, motion, implants, claustrophobia, contrast considerations and sequence/scanner dependence; incidental findings.

Reference and validation strategy. Phantom QA, harmonized protocols, test-retest, segmentation and clinical/pathologic reference.

Clinical/research maturity. Mature; quantitative biomarkers need harmonization.

Evidence anchors: [7,45]

10.0.8 Functional MRI

Measured quantity and transducer. Blood-oxygen-level dependent or perfusion signal as an indirect proxy for neural activity.

Typical outputs. Task/rest network maps, presurgical language/motor localization and research phenotypes.

Strengths. Whole-brain spatial mapping.

Limitations and artifacts. Hemodynamic rather than direct neural signal; motion, physiology, analytic flexibility and poor individual transport.

Reference and validation strategy. Physiologic monitoring, preregistration, test-retest, task validation and clinical outcome studies.

Clinical/research maturity. Mature research; selected clinical use.

Evidence anchors: [10,45]

10.0.9 PET and SPECT

Measured quantity and transducer. Radiotracers measure metabolism, perfusion, transporters, receptors or pathologic proteins.

Typical outputs. Tumor, seizure focus, neurodegeneration biomarkers and selected movement/cognitive differentials.

Strengths. Molecular specificity unavailable to structural imaging.

Limitations and artifacts. Radiation, cost, tracer specificity, partial-volume effects and biomarker-disease ambiguity.

Reference and validation strategy. Tracer chemistry/QC, scanner calibration, pathology or longitudinal reference and utility studies.

Clinical/research maturity. Established specialized use.

Evidence anchors: [6,31]

10.0.10 Transcranial and carotid Doppler ultrasound

Measured quantity and transducer. Ultrasound frequency shifts estimate blood velocity in accessible vessels.

Typical outputs. Flow velocity, stenosis proxies, emboli, vasospasm and dynamic autoregulation.

Strengths. Portable, repeated and high temporal resolution.

Limitations and artifacts. Operator/window dependence; velocity is not flow without geometry; extracranial/intracranial coverage limits.

Reference and validation strategy. Flow phantoms, angle/protocol control, cross-imaging validation and repeatability.

Clinical/research maturity. Mature specialized method.

Evidence anchors: [7,24]

10.0.11 Lumbar puncture and CSF analysis

Measured quantity and transducer. Pressure measurement and biochemical, cellular, microbiologic, immunologic or molecular assays of CSF.

Typical outputs. Infection, inflammation, hemorrhage, pressure and selected neurodegenerative biomarkers.

Strengths. Samples a compartment close to CNS biology.

Limitations and artifacts. Headache, bleeding/infection risk, pressure-position effects, contamination, preanalytic variability and nonspecific biomarkers.

Reference and validation strategy. Manometer/assay QC, handling protocol, paired controls and clinical/pathologic validation.

Clinical/research maturity. Mature for defined indications.

Evidence anchors: [6,29]

10.0.12 Intracranial pressure and multimodal neurocritical monitoring

Measured quantity and transducer. Ventricular/parenchymal pressure sensors plus optional oxygen, flow, EEG and metabolism probes.

Typical outputs. ICP, CPP and local physiologic trends.

Strengths. Continuous data in severe injury and ability to drain CSF with ventricular systems.

Limitations and artifacts. Infection/bleeding, calibration drift, local sampling and uncertain treatment thresholds across phenotypes.

Reference and validation strategy. Bench pressure calibration, insertion/zeroing protocol, cross-sensor comparison and outcome-based protocols.

Clinical/research maturity. Mature in selected critical care.

Evidence anchors: [24,26]

10.0.13 Cognitive and neuropsychological assessment

Measured quantity and transducer. Standardized tasks measure attention, memory, language, executive, visuospatial and processing domains.

Typical outputs. Domain profile, validity, functional implications and longitudinal change.

Strengths. Directly assesses performance relevant to cognition.

Limitations and artifacts. Education, language, culture, sensory/motor limitations, effort, practice effects and ecological gap.

Reference and validation strategy. Appropriate norms, alternate forms, reliable change, accommodations and functional anchors.

Clinical/research maturity. Mature when properly standardized.

Evidence anchors: [10,39,41]

10.0.14 Wearable movement, gait, tremor and sleep sensing

Measured quantity and transducer. Inertial, pressure, optical, location and interaction sensors quantify movement and context.

Typical outputs. Activity, gait features, tremor, falls, medication cycles and selected sleep proxies.

Strengths. Long-duration real-world sampling.

Limitations and artifacts. Device placement, adherence, algorithm opacity, environmental confounding and inability to infer cause.

Reference and validation strategy. Reference motion capture/clinical scales, free-living validation, firmware tracking and subgroup performance.

Clinical/research maturity. Established narrow metrics to frontier composite biomarkers.

Evidence anchors: [22,45]

10.0.15 Eye tracking, pupillometry and oculography

Measured quantity and transducer. Camera or electro-oculography captures gaze, saccades, nystagmus and pupil dynamics.

Typical outputs. Cranial/brainstem/cerebellar function, attention, autonomic and fatigue proxies.

Strengths. Sensitive high-frequency motor output and bedside portability.

Limitations and artifacts. Lighting, glasses, eyelid, pigmentation/iris characteristics, calibration and drug effects.

Reference and validation strategy. Optical targets, known trajectories, repeatability and diagnosis-specific validation.

Clinical/research maturity. Established specialist and research tool.

Evidence anchors: ^[22]

10.0.16 Transcranial magnetic/electrical stimulation

Measured quantity and transducer. Magnetic or electrical fields perturb cortical excitability; EMG/EEG or behavior measures response.

Typical outputs. Motor thresholds, inhibition/facilitation, causal circuit tests and treatment delivery.

Strengths. Can probe and modulate circuits without surgery.

Limitations and artifacts. Coarse fields, state dependence, discomfort, seizure risk and variable individual targeting.

Reference and validation strategy. Field modeling, output calibration, safety screening, sham/blinding and clinical outcome trials.

Clinical/research maturity. Established for selected therapies; research elsewhere.

Evidence anchors: ^[20,24]

10.0.17 Implanted neurostimulation and brain-computer interfaces

Measured quantity and transducer. Deep, cortical, spinal or peripheral electrodes record and/or stimulate neural tissue.

Typical outputs. Symptom control, closed-loop biomarkers, communication or motor control.

Strengths. Direct access and potential closed-loop adaptation.

Limitations and artifacts. Surgery, infection, lead failure, tissue response, battery/cybersecurity, long-term data rights and dependence.

Reference and validation strategy. Bench/animal safety, human factors, locked algorithms, longitudinal device and functional outcomes, postmarket surveillance.

Clinical/research maturity. Mature for selected stimulation; BCI restoration remains emerging.

Evidence anchors: [20,23,24]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise nervous system health and neurotechnology quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Continued on next page

Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

11.0.1 Intended use and measurand

Design requirements. Define the precise nervous system health and neurotechnology quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: ^[22]

11.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: ^[24]

11.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

11.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

11.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

11.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: [45]

11.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [21,45]

11.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [23,24]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Vascular risk prevention	Primary and secondary prevention	Control pressure, tobacco, diabetes, lipids, atrial fibrillation and activity according to absolute risk and prior events.	Strong causal evidence for preventing stroke and vascular cognitive injury when implemented appropriately.	Medication adverse effects, polypharmacy, inequitable access and overreliance on scores.	Risk factors, adherence, bleeding/falls, function and recurrent events.
Time-critical reperfusion and hemorrhage care	Emergency system and procedure	Restore flow in eligible ischemia or control bleeding/pressure in hemorrhage through coordinated imaging, drugs and procedures.	Large time-dependent benefit in selected acute stroke populations.	Hemorrhage, reperfusion injury, procedure complications, transfer delay and selection inequity.	Time metrics, imaging/neurologic response, hemorrhage, disability and system reach.
Antiseizure medication and rescue therapy	Pharmacology	Reduce network excitability or interrupt prolonged seizures through channel, receptor or synaptic targets.	Effective for many seizure types; selection depends on syndrome, comorbidity, pregnancy potential and interactions.	Sedation, mood/cognition, rash, teratogenicity, organ toxicity and withdrawal seizures.	Seizure diary, levels selectively, adverse effects, interactions, pregnancy planning and quality of life.
Disease-modifying immunotherapy	Pharmacology	Supplements, redirect or replace immune pathways in demyelinating, neuromuscular or autoimmune neurologic disease.	Substantial benefit in defined diseases and activity states; timing matters.	Infection, malignancy, infusion reactions, reproductive issues, cost and rebound after cessation.	Clinical/MRI activity, blood counts/organ function, infection screening, vaccination and treatment-specific markers.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Neurodegenerative symptomatic and targeted therapy	Pharmacology/Molecular	Modulate neurotransmission, protein pathways or gene/RNA mechanisms to improve symptoms or slow selected disease.	Symptomatic benefits are established for several disorders; disease modification is modest or subtype/stage restricted.	Neuropsychiatric, cardiovascular, dyskinesia, edema/hemorrhage imaging findings, infusion burden and uncertain long-term value.	Function and goals, adverse effects, imaging/laboratory requirements, caregiver burden and discontinuation rules.
Neurosurgery and endovascular procedures	Procedure	Remove, decompress, repair, revascularize, ablate or sample structural disease.	Can be lifesaving or definitive when lesion, timing and patient goals align.	Bleeding, infection, deficit, anesthesia, device failure and irreversible tradeoffs.	Neurologic/function outcome, imaging, pathology, complications and patient-reported goals.
Neurostimulation and neuromodulation	Device therapy	Deliver electrical, magnetic or other energy to alter circuits in movement, epilepsy, pain, mood or rehabilitation.	Effective for selected indications; closed-loop systems can tailor stimulation to signals.	Surgery or seizure risk, cognition/mood effects, hardware failure, programming burden and cybersecurity.	Target symptoms, adverse effects, device integrity, battery, stimulation logs, function and patient control.
Physical, occupational, speech and cognitive rehabilitation	Rehabilitation	Use task-specific practice, compensation, environment, assistive technology and education to improve activity and participation.	Improves outcomes after stroke, injury and many chronic disorders when dose and goals are appropriate.	Fatigue, access, inadequate intensity, overpromised technology and mismatch with home context.	Goal attainment, task performance, participation, burden, caregiver outcomes and retention.
Pain and symptom management	Multimodal supportive care	Combine mechanism-informed medication, physical, psychological, behavioral and interventional approaches.	Some therapies provide modest domain-specific benefit; coordinated care is preferable to single-modality escalation.	Sedation, dependence, procedure harm, stigma, cost and undertreatment/overmedicalization.	Pain interference, function, mood/sleep, adverse effects, dose and agreed goals.
Sleep and circadian intervention	Behavioral/systemic	Treat sleep apnea, insomnia, circadian disruption, restless legs and medication/environmental contributors.	Sleep treatment can improve symptoms and safety; neurologic disease-specific outcome effects vary.	Sedation, falls, complex behaviors, device nonadherence and circadian mismatch.	Sleep timing/quality, daytime function, apnea metrics where relevant, medication effects and safety.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Nutrition, swallowing and respiratory support	Supportive/assistive	Maintain safe intake, energy, secretion clearance, ventilation and communication in bulbar or neuromuscular disease.	Improves safety, comfort and survival in selected progressive disease when aligned with goals.	Aspiration, procedure burden, loss of autonomy, device complications and caregiver load.	Weight/nutrition swallow safety, respiratory measures, communication, goals and complications.
Psychological, social and caregiver support	Whole-person care	Address adjustment, depression/anxiety, cognition, family roles, employment, stigma and caregiver strain.	Improves engagement and quality; social determinants strongly shape access and outcome.	Fragmentation, stigma, cultural mismatch and shifting burden to family.	Patient/caregiver reported outcomes, participation, crisis risk, service access and equity.

12.0.1 Vascular risk prevention

Target and mechanism. Control pressure, tobacco, diabetes, lipids, atrial fibrillation and activity according to absolute risk and prior events.

Evidence and expected outcome. Strong causal evidence for preventing stroke and vascular cognitive injury when implemented appropriately.

Risks, burden, and equity. Medication adverse effects, polypharmacy, inequitable access and overreliance on scores.

Response monitoring. Risk factors, adherence, bleeding/falls, function and recurrent events.

Evidence anchors: [2,7]

12.0.2 Time-critical reperfusion and hemorrhage care

Target and mechanism. Restore flow in eligible ischemia or control bleeding/pressure in hemorrhage through coordinated imaging, drugs and procedures.

Evidence and expected outcome. Large time-dependent benefit in selected acute stroke populations.

Risks, burden, and equity. Hemorrhage, reperfusion injury, procedure complications, transfer delay and selection inequity.

Response monitoring. Time metrics, imaging/neurologic response, hemorrhage, disability and system reach.

Evidence anchors: [7,26]

12.0.3 Antiseizure medication and rescue therapy

Target and mechanism. Reduce network excitability or interrupt prolonged seizures through channel, receptor or synaptic targets.

Evidence and expected outcome. Effective for many seizure types; selection depends on syndrome, comorbidity, pregnancy potential and interactions.

Risks, burden, and equity. Sedation, mood/cognition, rash, teratogenicity, organ toxicity and withdrawal seizures.

Response monitoring. Seizure diary, levels selectively, adverse effects, interactions, pregnancy planning and quality of life.

Evidence anchors: [8,25]

12.0.4 Disease-modifying immunotherapy

Target and mechanism. Suppress, redirect or replace immune pathways in demyelinating, neuromuscular or autoimmune neurologic disease.

Evidence and expected outcome. Substantial benefit in defined diseases and activity states; timing matters.

Risks, burden, and equity. Infection, malignancy, infusion reactions, reproductive issues, cost and rebound after cessation.

Response monitoring. Clinical/MRI activity, blood counts/organ function, infection screening, vaccination and treatment-specific markers.

Evidence anchors: [6,34]

12.0.5 Neurodegenerative symptomatic and targeted therapy

Target and mechanism. Modulate neurotransmission, protein pathways or gene/RNA mechanisms to improve symptoms or slow selected disease.

Evidence and expected outcome. Symptomatic benefits are established for several disorders; disease modification is modest or subtype/stage restricted.

Risks, burden, and equity. Neuropsychiatric, cardiovascular, dyskinesia, edema/hemorrhage imaging findings, infusion burden and uncertain long-term value.

Response monitoring. Function and goals, adverse effects, imaging/laboratory requirements, caregiver burden and discontinuation rules.

Evidence anchors: [6,9,14]

12.0.6 Neurosurgery and endovascular procedures

Target and mechanism. Remove, decompress, repair, revascularize, ablate or sample structural disease.

Evidence and expected outcome. Can be lifesaving or definitive when lesion, timing and patient goals align.

Risks, burden, and equity. Bleeding, infection, deficit, anesthesia, device failure and irreversible trade-offs.

Response monitoring. Neurologic/functional outcome, imaging, pathology, complications and patient-reported goals.

Evidence anchors: [7,26,31]

12.0.7 Neurostimulation and neuromodulation

Target and mechanism. Deliver electrical, magnetic or other energy to alter circuits in movement, epilepsy, pain, mood or rehabilitation.

Evidence and expected outcome. Effective for selected indications; closed-loop systems can tailor stimulation to signals.

Risks, burden, and equity. Surgery or seizure risk, cognition/mood effects, hardware failure, programming burden and cybersecurity.

Response monitoring. Target symptoms, adverse effects, device integrity, battery, stimulation logs, function and patient control.

Evidence anchors: [20,23,24]

12.0.8 Physical, occupational, speech and cognitive rehabilitation

Target and mechanism. Use task-specific practice, compensation, environment, assistive technology and education to improve activity and participation.

Evidence and expected outcome. Improves outcomes after stroke, injury and many chronic disorders when dose and goals are appropriate.

Risks, burden, and equity. Fatigue, access, inadequate intensity, overpromised technology and mismatch with home context.

Response monitoring. Goal attainment, task performance, participation, burden, caregiver outcomes and retention.

Evidence anchors: [16,17,18]

12.0.9 Pain and symptom management

Target and mechanism. Combine mechanism-informed medication, physical, psychological, behavioral and interventional approaches.

Evidence and expected outcome. Some therapies provide modest domain-specific benefit; coordinated care is preferable to single-modality escalation.

Risks, burden, and equity. Sedation, dependence, procedure harm, stigma, cost and undertreatment/overmedicalization.

Response monitoring. Pain interference, function, mood/sleep, adverse effects, dose and agreed goals.

Evidence anchors: [27,42]

12.0.10 Sleep and circadian intervention

Target and mechanism. Treat sleep apnea, insomnia, circadian disruption, restless legs and medication/environmental contributors.

Evidence and expected outcome. Sleep treatment can improve symptoms and safety; neurologic disease-specific outcome effects vary.

Risks, burden, and equity. Sedation, falls, complex behaviors, device nonadherence and circadian mismatch.

Response monitoring. Sleep timing/quality, daytime function, apnea metrics where relevant, medication effects and safety.

Evidence anchors: [22,28]

12.0.11 Nutrition, swallowing and respiratory support

Target and mechanism. Maintain safe intake, energy, secretion clearance, ventilation and communication in bulbar or neuromuscular disease.

Evidence and expected outcome. Improves safety, comfort and survival in selected progressive disease when aligned with goals.

Risks, burden, and equity. Aspiration, procedure burden, loss of autonomy, device complications and caregiver load.

Response monitoring. Weight/nutrition, swallow safety, respiratory measures, communication, goals and complications.

Evidence anchors: [16,18,42]

12.0.12 Psychological, social and caregiver support

Target and mechanism. Address adjustment, depression/anxiety, cognition, family roles, employment, stigma and caregiver strain.

Evidence and expected outcome. Improves engagement and quality; social determinants strongly shape access and outcome.

Risks, burden, and equity. Fragmentation, stigma, cultural mismatch and shifting burden to family.

Response monitoring. Patient/caregiver reported outcomes, participation, crisis risk, service access and equity.

Evidence anchors: [38,39,40]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can digital motor and cognitive biomarkers replace clinic-based outcomes?	Wearables and tasks can quantify selected behaviors frequently and objectively.	Whether they are meaningful, stable, interpretable and treatment-responsive across devices, homes and disability states.	Behavior is context-dependent; missingness and adaptation may be informative; proprietary algorithms change.	Prospective external validation anchored to patient-important function, with raw data, context and version control.
How should preclinical neurodegenerative biomarkers be used?	Molecular and imaging abnormalities can precede symptoms and enrich risk.	Net benefit of disclosure, surveillance or treatment for asymptomatic individuals across risk levels.	Penetrance is incomplete; anxiety, insurance/social harms, overdiagnosis and uncertain intervention benefit matter.	Randomized disclosure/intervention studies with long follow-up, rights protections and functional outcomes.
Can closed-loop stimulation discover individualized causal circuits?	Adaptive systems can link sensed states to stimulation and may improve efficiency.	Whether learned biomarkers are causal, transportable and stable over years and disease progression.	The controller changes the system it observes; reinforcement and drift can produce hidden failure.	Prospective system-identification protocols, sham/crossover phases, interpretable logs and long-term safety endpoints.
What drives persistent symptoms after mild neurologic injury or infection?	Sleep, autonomic, vestibular, cognitive, inflammatory, pain and psychosocial mechanisms can contribute.	Reliable subtypes and matching treatments.	Broad labels mix mechanisms; expectation, compensation and concurrent disease complicate causal inference.	Longitudinal multimodal phenotyping followed by mechanism-targeted randomized trials.

Continued on next page

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within nervous system health and neurotechnology.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

13.0.1 Can digital motor and cognitive biomarkers replace clinic-based outcomes?

Established knowledge. Wearables and tasks can quantify selected behaviors frequently and objectively.

Unresolved issue. Whether they are meaningful, stable, interpretable and treatment-responsive across devices, homes and disability states.

Why current evidence is insufficient. Behavior is context-dependent; missingness and adaptation may be informative; proprietary algorithms change.

Research that would change confidence. Prospective external validation anchored to patient-important function, with raw data, context and version control.

Evidence anchors: [22,41,45]

13.0.2 How should preclinical neurodegenerative biomarkers be used?

Established knowledge. Molecular and imaging abnormalities can precede symptoms and enrich risk.

Unresolved issue. Net benefit of disclosure, surveillance or treatment for asymptomatic individuals across risk levels.

Why current evidence is insufficient. Penetrance is incomplete; anxiety, insurance/social harms, over-diagnosis and uncertain intervention benefit matter.

Research that would change confidence. Randomized disclosure/intervention studies with long follow-up, rights protections and functional outcomes.

Evidence anchors: [14,20,44]

13.0.3 Can closed-loop stimulation discover individualized causal circuits?

Established knowledge. Adaptive systems can link sensed states to stimulation and may improve efficiency.

Unresolved issue. Whether learned biomarkers are causal, transportable and stable over years and disease progression.

Why current evidence is insufficient. The controller changes the system it observes; reinforcement and drift can produce hidden failure.

Research that would change confidence. Prospective system-identification protocols, sham/crossover phases, interpretable logs and long-term safety endpoints.

Evidence anchors: [21,23,46]

13.0.4 What drives persistent symptoms after mild neurologic injury or infection?

Established knowledge. Sleep, autonomic, vestibular, cognitive, inflammatory, pain and psychosocial mechanisms can contribute.

Unresolved issue. Reliable subtypes and matching treatments.

Why current evidence is insufficient. Broad labels mix mechanisms; expectation, compensation and concurrent disease complicate causal inference.

Research that would change confidence. Longitudinal multimodal phenotyping followed by mechanism-targeted randomized trials.

Evidence anchors: [10,16,43]

13.0.5 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within nervous system health and neurotechnology.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: [14,43]

13.0.6 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: [41,45]

13.0.7 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[31,44]

13.0.8 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[22]

13.0.9 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[17,41]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Multimodal event truth set	Can synchronized EEG, video, inertial, ECG and clinician annotation distinguish seizures, syncope, sleep events and functional attacks?	Prospective multicenter capture of typical events with blinded adjudication and locked algorithms; preserve all failed recordings.	Calibrated event-level performance and abstention across sites, ages and event classes with clinical utility.	High accuracy depends on leakage, curated clips or excluding ambiguous events.
0. Neurodevice continuity study	What happens to implanted-system users across hardware failure, software updates and vendor discontinuation?	International registry with device versions, security events, support access, explants, function, agency and cost.	Actionable standards for continuity and no subgroup abandonment.	Only technical performance reported while dependence and loss of service remain invisible.
0. Recovery reserve model	Can early physiology and functional testing predict recovery without creating self-fulfilling treatment withdrawal?	Prospective cohort with blinded prognostic model, repeated assessments, treatment decisions recorded and counterfactual sensitivity analysis.	Well-calibrated ranges that add value while preserving uncertainty and avoiding treatment-mediated bias.	Model reflects clinician decisions more than biology or worsens inequitable withdrawal.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for nervous system health and neurotechnology?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.

Continued on next page

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

14.0.1 0. Multimodal event truth set

Question. Can synchronized EEG, video, inertial, ECG and clinician annotation distinguish seizures, syncope, sleep events and functional attacks?

Design. Prospective multicenter capture of typical events with blinded adjudication and locked algorithms; preserve all failed recordings.

Success criterion. Calibrated event-level performance and abstention across sites, ages and event classes with clinical utility.

Failure or stop criterion. High accuracy depends on leakage, curated clips or excluding ambiguous events.

Evidence anchors: [8,45]

14.0.2 0. Neurodevice continuity study

Question. What happens to implanted-system users across hardware failure, software updates and vendor discontinuation?

Design. International registry with device versions, security events, support access, explants, function, agency and cost.

Success criterion. Actionable standards for continuity and no subgroup abandonment.

Failure or stop criterion. Only technical performance reported while dependence and loss of service remain invisible.

Evidence anchors: [19,20,23]

14.0.3 0. Recovery reserve model

Question. Can early physiology and functional testing predict recovery without creating self-fulfilling treatment withdrawal?

Design. Prospective cohort with blinded prognostic model, repeated assessments, treatment decisions recorded and counterfactual sensitivity analysis.

Success criterion. Well-calibrated ranges that add value while preserving uncertainty and avoiding treatment-mediated bias.

Failure or stop criterion. Model reflects clinician decisions more than biology or worsens inequitable withdrawal.

Evidence anchors: [26,40,43]

14.0.4 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for nervous system health and neurotechnology?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: [40]

14.0.5 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

Evidence anchors: ^[43]

14.0.6 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[45]

14.0.7 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: ^[44]

14.0.8 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: ^[17]

14.0.9 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[40]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03A	Cardiovascular Health	Cerebral perfusion, embolism, autonomic control and vascular prevention.	stroke, syncope, cognitive decline or orthostatic symptoms are evaluated.
03B	Respiratory Health	Control of breathing, sleep, bulbar protection and neuromuscular ventilatory failure.	brainstem, spinal, nerve or muscle disease affects breathing.
03D	Endocrine and Metabolic Health	Glucose, thyroid, adrenal, electrolyte and mitochondrial effects on neural function.	encephalopathy, neuropathy or episodic dysfunction may be metabolic.
03I	Musculoskeletal Health	Motor output, biomechanics, pain, gait and rehabilitation.	weakness or movement limitation must be localized.
03K	Sensory Health	Vision, hearing and vestibular inputs to cognition, balance and communication.	sensory loss or pathway disease alters the phenotype.
05	Brain, Cognition, Emotion and Behavior	Dimensional mental, cognitive and behavioral frameworks across diagnoses.	symptoms cannot be understood through lesion localization alone.
10	Function, Disability and Rehabilitation	Activity, participation, assistive technology and recovery.	outcomes and goals extend beyond impairment.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Aphasia	Acquired impairment of language processing, distinguished from dysarthria and cognitive or sensory barriers.
Electroencephalogram	Scalp measurement of time-varying electrical fields generated mainly by synchronized postsynaptic currents.
Localization	Inference from signs and tests to the level or network in the nervous system likely to be affected.
Neuroplasticity	Experience- or injury-related change in neural function or structure; it can be adaptive or maladaptive.
Seizure	Transient signs or symptoms due to abnormal excessive or synchronous neuronal activity.
Upper motor neuron	Descending motor pathways in brain and spinal cord whose dysfunction produces characteristic weakness, tone and reflex patterns.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

- [1] Technical Evidence and Systems Review. Atlas of Human Health: A Unified Taxonomy. 2026. Master taxonomy and conceptual architecture for this suite.
- [2] Technical Evidence and Engineering Review. Cardiovascular Health and Measurement Engineering. 2026. Volume 03A of this suite.
- [3] Alberts B, Johnson A, Lewis J, et al.. Molecular Biology of the Cell, 7th ed.. 2022.
- [4] Boron WF, Boulpaep EL. Medical Physiology, 4th ed.. 2022.
- [5] Kumar V, Abbas AK, Aster JC. Robbins and Cotran Pathologic Basis of Disease, 11th ed.. 2025.
- [6] American Academy of Neurology. Clinical practice guidelines. 2026. <https://www.aan.com/Guidelines/home>
- [7] American Heart Association/American Stroke Association. Guidelines for the prevention and management of stroke. 2024-2026. <https://professional.heart.org/en/guidelines-and-statements>
- [8] International League Against Epilepsy. Epilepsy classification and clinical standards. 2025/2026. <https://www.ilae.org/guidelines>
- [9] Movement Disorder Society. Clinical resources and rating scales. 2026. <https://www.movementdisorders.org/MDS/Resources/Clinical-Resources.htm>
- [10] National Institute of Mental Health. Research Domain Criteria (RDoC). 2026. <https://www.nimh.nih.gov/research/research-funded-by-nimh/rdoc>
- [11] Kotov R, Krueger RF, Watson D, et al.. The Hierarchical Taxonomy of Psychopathology: a dimensional alternative to traditional nosologies. 2017. <https://doi.org/10.1037/abn0000258>
- [12] American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision. 2022/2025. <https://www.psychiatry.org/psychiatrists/practice/dsm>
- [13] Human Cell Atlas Consortium. The Human Cell Atlas: towards a first draft atlas. 2024. <https://www.nature.com/collections/jccbbdahji>
- [14] World Health Organization. Precision medicine: a path towards targeted, personalized and equitable care (WHA79.8). 2026. <https://www.who.int/news/item/22-05-2026-world-health-assembly-endorses>
- [15] Lopez-Otin C, Blasco MA, Partridge L, Serrano M, Kroemer G. Hallmarks of aging: an expanding universe. 2023. <https://doi.org/10.1016/j.cell.2022.11.001>
- [16] World Health Organization. Rehabilitation 2030 initiative. 2026. <https://www.who.int/initiatives/rehabilitation-2030>
- [17] World Health Organization. International Classification of Functioning, Disability and Health. 2001/2026. <https://www.who.int/standards/classifications/international-classification-of-functioning>
- [18] World Health Organization and UNICEF. Global report on assistive technology. 2022/2026. <https://www.who.int/publications/i/item/9789240049451>

- [19] United Nations. Convention on the Rights of Persons with Disabilities. 2006. <https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-persons-disabilities>
- [20] UNESCO. Recommendation on the Ethics of Neurotechnology. 2025. <https://www.unesco.org/en/legal-affairs/recommendation-ethics-neurotechnology>
- [21] U.S. Food and Drug Administration. Artificial Intelligence-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations. 2025. <https://www.fda.gov/media/184856/download>
- [22] U.S. Food and Drug Administration. Digital Health Center of Excellence and current guidance. 2026. <https://www.fda.gov/medical-devices/digital-health-center-excellence>
- [23] U.S. Food and Drug Administration. Cybersecurity in Medical Devices: Quality Management System Considerations and Content of Premarket Submissions. 2026. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/cybersecurity-medical-devices-quality-management-system-con>
- [24] International Electrotechnical Commission. IEC 60601-1: Medical electrical equipment - basic safety and essential performance. 2005+A1+A2. <https://webstore.iec.ch/publication/67497>
- [25] International Union of Basic and Clinical Pharmacology. IUPHAR/BPS Guide to Pharmacology. 2026. <https://www.guidetopharmacology.org/>
- [26] World Health Organization. Global Patient Safety Action Plan 2021-2030. 2021. <https://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan>
- [27] International Association for the Study of Pain. IASP terminology and ICD-11 pain classification. 2025/2026. <https://www.iasp-pain.org/resources/terminology/>
- [28] American Academy of Sleep Medicine. Clinical practice guidelines and scoring manual. 2025/2026. <https://aasm.org/clinical-resources/practice-standards/practice-guidelines/>
- [29] National Institute of Allergy and Infectious Diseases. Research priorities in infection, immunity, and microbiome. 2025/2026. <https://www.niaid.nih.gov/research>
- [30] World Health Organization. Global Action Plan on Antimicrobial Resistance 2026-2036. 2026. <https://www.who.int/news/item/25-05-2026-the-world-health-assembly-adopts-updated-global-action-p-282026-2036%29>
- [31] National Cancer Institute. Cancer biology, screening, diagnosis and treatment resources. 2026. <https://www.cancer.gov/>
- [32] Hanahan D. Hallmarks of Cancer: New Dimensions. 2022. <https://doi.org/10.1158/2159-8290.CD-21-1059>
- [33] American Society of Hematology. ASH clinical practice guidelines. 2026. <https://www.hematology.org/education/clinicians/guidelines-and-quality-care/clinical-practice-guidelines>
- [34] American College of Rheumatology. Clinical practice guidelines. 2026. <https://rheumatology.org/clinical-practice-guidelines>
- [35] American Diabetes Association. Standards of Care in Diabetes - 2026. 2026. https://diabetesjournals.org/care/issue/49/Supplement_1
- [36] Barker DJP. The developmental origins of adult disease. 2004. <https://doi.org/10.1080/08035320410022730>
- [37] World Health Organization. Life course approach to health. 2025/2026. <https://www.who.int/our-work/life-course>
- [38] World Health Organization Commission on Social Connection. From loneliness to social connection: charting a path to healthier societies. 2025. <https://www.who.int/publications/i/item/9789240112681>

- [39] World Health Organization. WHOQOL: Measuring Quality of Life. 2026. <https://www.who.int/tools/whoqol>
- [40] World Health Organization. Health equity. 2026. <https://www.who.int/health-topics/health-equity>
- [41] National Institutes of Health. Patient-Reported Outcomes Measurement Information System (PROMIS). 2026. <https://commonfund.nih.gov/promis>
- [42] World Health Organization. Palliative care. 2025. <https://www.who.int/news-room/fact-sheets/detail/palliative-care>
- [43] Hernan MA, Robins JM. Causal Inference: What If. 2020. <https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/>
- [44] Guyatt GH, Oxman AD, Vist GE, et al.. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. 2008. <https://doi.org/10.1136/bmj.39489.470347.AD>
- [45] Collins GS, Moons KGM, Dhiman P, et al.. TRIPOD+AI statement. 2024. <https://www.tripod-statement.org/>
- [46] Kitano H. Systems biology: a brief overview. 2002. <https://doi.org/10.1126/science.1069492>

Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.